

NetForensics — Pipeline Flow

Explainable Packet Forensics with Court-Admissible Evidence Handling

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INPUT

.pcap capture file

A recording of network traffic from an incident — the raw evidence under investigation.



STAGE 1

Ingest & Seal

EVIDENCE INTEGRITY

Before any analysis: hash the file with **BLAKE3** and record it with an **RFC 3161 trusted timestamp**. Establishes “this exact evidence, unaltered, received at this time.” The digital evidence-bag seal.



STAGE 2

Parse to Flows

dpkt for the fast bulk pass; **PyShark / tshark** (Wireshark dissectors) for protocol-rich TLS & DNS fields. Output: connection flows with extracted features — counts, timing, DNS queries, TLS handshakes.



STAGE 3

Run Explainable Detectors

Each detector emits a finding *with the evidence that triggered it* — no black boxes.

DNS exfiltration / tunneling

Shannon entropy + subdomain length + query rate → encoded-data randomness

C2 beaconing

Inter-arrival-time regularity → robotic, too-regular check-ins

Port scan / recon

Fan-out analysis → one source touching many ports/hosts fast

Malicious TLS client

JA3 fingerprint matched vs. abuse.ch SSLBL blocklist



STAGE 4

Build the Case File

EVIDENCE INTEGRITY

Findings + evidence + integrity record → **append-only custody log** → **forensic PDF report**. Plain-language reasoning a prosecutor or magistrate can follow, with a verifiable chain of custody from ingest to report.

Design principle: explainability and evidentiary integrity over algorithmic sophistication. Every finding must be understandable, defensible, and court-admissible.

The differentiator (shaded / sealed stages): existing tools analyse traffic; NetForensics packages analysis into court-admissible evidence with a tamper-evident chain of custody.